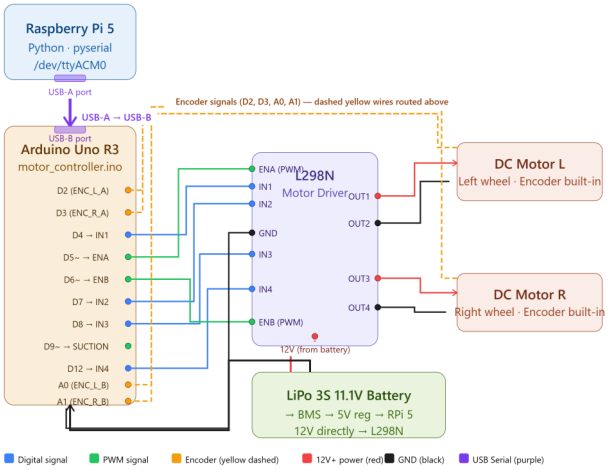
University of Hertfordshire UH

Logbook 4

Name:	**Jovian Lim Zheng Yuan**
Week Date:	**Part A -Meeting 4 of 8**

*(The above information is for you to keep track of which week this log is applicable for. This information is not required in the online logbook as the system will find your name, your supervisor's name, and take the submission date, automatically. You must submit the online log in the relevant week. Logs **cannot** be backdated to missed weeks.)*

Meeting date: *(please note if no meeting is planned this week)*	**Date and day** **23 July 2026, Thursday**
Work Undertaken this week: *Not more than 500 words*	**From the last meeting until now, what have you done based on the task you have listed in the last logbook?** I have completed the draft project outline and put together the draft project resource list, with the contents added in. Next, I tested a simple Python code across several files in one folder and confirmed it runs smoothly. Did research on my parts weight and dimensions, after considerations the estimated weight after its assembled is 1.5kg and after the investigations heaviest part is the chassis with all the controls. Did brief wiring connections but will improved it in project outline. Following up on the raspberry pi and Arduino link it uses a USB cable with UART pins to communicate with each other Lastly, I did research on CNN for dust/dirt detection and floor type detection I will be using MobileNetV2 for floor classification and a patch-based binary CNN for dirt detection.
Problems encountered: *(Not more than 500 words*	**What are the tasks that are still outstanding to write your Project Outline** Finalize the project outline by adding wiring diagram and system flowchart. Add in the 3D drawing of my model. Fix gantt chart plus add an updated BOM list. Update the risk assessment. Lastly, do up an improved wiring diagram for the project outline
Tasks agreed for upcoming week before the next meeting	**What are your plan actions/task until the next progress meeting that are agreed with your supervisor** **1. Firstly, finished up with the project outline and if possible do up the 3d drawing of the full scale vacuum cleaner.** **2. Finalize the BOM list for project outline** **3. Next, research more in depth how the CNN applications works after finding out what it does, like what was mentioned in work undertaken.** **4. Design and plan the chassis of the model and as well as the impeller using fusion 360 or other cad software.** **5. An app to monitor robot cleaning progress has yet**

	to fulfill and developed **6. Further do more investigation on baby ros with python raspberry pi integration** **7. Investigate the components lead time, better flow of preparations** **8. Test on the key codes on how they work together**
Figures: *(Upload image files in the online Log Form, but feel free to keep notes here about which images you intend to upload. Use "Figure Caption" information as the filenames, e.g., "Figure 1 – A screenshot of the initial constrained Sketch in CATIA.jpg")*	**1) Update Hardware Block diagram of your system**  Figure 1: System Wiring Diagram

System flowchart/ architecture

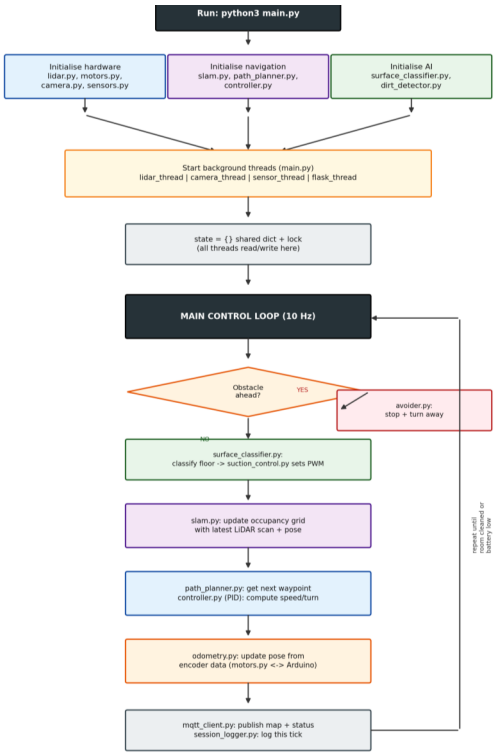


Figure 2: system flowchart from top to bottom

2) Key software codes/modules that need to be developed

There will be multiple codes to integrate into this project like such as to call out the hardware, navigation, ai logical thinking and for app communication plus some are utilities code for the backend.

have split them to groups

- Hardware - lidar.py its to call the lidar out to get the distance and views - motor.py control the motor speed and direction -camera.py calls the camera for use in CNN output -beacon.py for homing when it needs to recharge or go home - Sensors.py reads the sensors, battery voltage, obstacle avoidance -odometry.py mainly for reading the encoder cycles and showing the axis of the robot
- Navigation -Slam.py updated the map from the lidar scanning plus Odmetry -path_planner.py it generates the path to clean from the map -Obstacle Avoidance.py for wall avoidance -controller.py to track the robot between point to point

- AI aspects
  -surface_classifier.py identify the tiles on the floor appearance like
  carpet or wood or others

-dirt_detector.py sees dirt/dust on maps and marks as dirt section
-suction_control.py controls the power of the motor based on the
floor type
-train_classifier.py trains the model

• App codes
- Server.py for map logging
- Mqtt_client.py for map updates
- Map-encoders to convert numpy to json file for display
- Session_logger.py save all logs and history of clean history
• Utilities
- Visualizer.py display the map on the app - Logger.py in the terminal
- Simulator.py testing the programme to see how the lidar works
with - Arduino_flasher.py communicate with raspberry

3) Any Quality and Regulatory issues addressed?

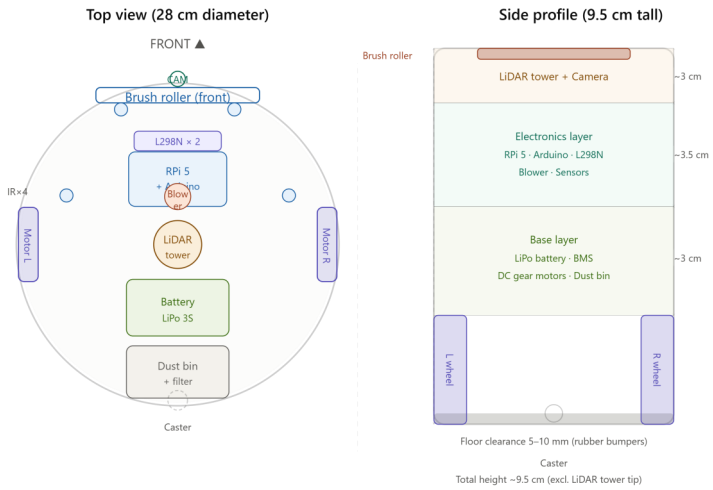
Quality:

1. Mechanical safety, since there are moving parts like brushes and a roller guard, so a finger or pet won't get caught in between.
2. Electrical safety to protect against short circuits, overcharging, and cell balancing protection for the BMS, proper wire gauges to prevent wire heating, and a flame-retardant LiPo battery casing so it can be kept safe.
3. The LiPo battery may overcharge, so adding a battery management system will cut the power before it overheats. Include temperature sensors to monitor the motor heat and shut it down when it is too hot.

Regulatory:

1. Product safety: since the vacuum uses mains power, a safety mark for consumer protection is needed.
2. Camera/data privacy: PDPA (Personal Data Protection Act 2012) the camera captures the house, so this must be addressed since it is used for mapping, and people may also get captured in the camera.
3. Battery Fire Safety Act SS 638 I will be using a LiPo battery, which needs to be properly labelled so it won't cause issues for others.

4) A updated/improved version of your system sketch

	 **Figure 3: system sketch with dimensions**
References:	**References used for quality and regulatory** **IEC 60335-2-2 — Household vacuum cleaners (electrical safety)** **IEC 62133 — Lithium battery safety** **ISO 10218 / ISO 13482 — Robotics safety (ISO 13482 covers personal care robots, very relevant for a home robot)** **FCC / CE marking — For Wi-Fi/Bluetooth modules used in the robot**